Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 2

September 18, 2026

Exercise 1. Which of the following represents the CNF and DNF forms of the compound proposition:

$$(p \vee \neg q) \wedge (\neg p \vee q)$$

- ◯ CNF: $(p \vee \neg q) \wedge (\neg p \vee q)$, DNF: $(p \vee \neg q) \wedge (\neg p \vee q)$
- ◯ CNF: $(p \vee \neg q) \wedge (\neg p \vee q)$, DNF: $(p \wedge q) \vee (\neg p \wedge \neg q)$
- ◯ CNF: $(p \vee q) \wedge (\neg p \vee \neg q)$, DNF: $(p \wedge q) \vee (\neg p \wedge \neg q)$
- ◯ CNF: $(p \wedge q) \vee (\neg p \wedge \neg q)$, DNF: $(p \vee q) \wedge (\neg p \vee \neg q)$

Exercise 2.

1. Find a DNF of $p \oplus (q \wedge r)$ without using a truth table.
2. Find a CNF of $[(p \to q) \vee (r \to s)] \wedge u$ without using a truth table.

Exercise 3. (Warm-up; unlike the multiple choice questions at the final exam, more than a single answer may be correct.). The law regarding alcohol drinking in Switzerland states the following: "If you are drinking strong alcohol, then you must be at least 18 years old." (for all x, $A(x) \to 18(x)$). Assume that you are hired as a police officer at Satellite and need to enforce the above law. Which student(s) are you going to arrest and why:

- ◯ *The 17-year old student drinking soda.*
- ◯ *The 17-year old student drinking vodka.*
- ◯ *The 18-year old student drinking water.*
- ◯ *The 18-year old student drinking tequila.*

Next Tuesday, Satellite organizes a cabaret night open to all students under the following conditions: "Everyone must be at least 18 years old and drink strong alcohol" (for all x, $A(x) \wedge 18(x)$). As previously, which student(s) are you going to arrest and why:

- ◯ *The 17-year old student drinking soda.*
- ◯ *The 17-year old student drinking vodka.*
- ◯ *The 18-year old student drinking water.*
- ◯ *The 18-year old student drinking tequila.*

Exercise 4. (Rosen, exercise 8, 1.4.14 in 8th edition) Translate these statements into English, where $R(x)$ is "x is a rabbit" and $H(x)$ is "x hops" and the domain consists of all animals.

1. $\forall x(R(x) \to H(x))$

2. $\exists x(R(x) \to H(x))$
3. $\forall x(R(x) \wedge H(x))$
4. $\exists x(R(x) \wedge H(x))$

Exercise 5. Which expressions below are equivalent to $\neg(\forall x \exists y\, P(x, y))$. Explain.

◯ $\exists x \forall y\ \neg P(x, y)$;

◯ $\exists x \exists y\ \neg P(x, y)$.

Exercise 6. (Rosen, exercise 9, 1.5.8 in 8th edition) Let $L(x, y)$ be the statement "x loves y", where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements:

1. Everybody loves Sharon.
2. Everybody loves somebody.
3. There is somebody whom everybody loves.
4. Nobody loves everybody.
5. There is somebody whom Daisy does not love.
6. There is somebody whom no one loves.
7. There is exactly one person whom everybody loves.
8. There are exactly two people whom Marsellus loves.
9. Everyone loves himself or herself.
10. There is someone who loves no one besides himself or herself.

Exercise 7. Given the two statements below, where the domain of discourse is $\mathbf{R}$ for both x and y,

$$\exists y \forall x(x \neq 0 \to xy = 1) \qquad\qquad \exists x \forall y(xy < 0 \to xy > 0)$$

◯ They are both false.

◯ Only the first is true.

◯ Only the second is true.

◯ They are both true.

Exercise 8. Consider the two statements below, where $P(x, y)$ is a propositional function and the domain of discourse is $\mathbf{Z}_{\geq 0}$ for x, y and z:

$$(\exists y \forall x\, P(x, y)) \ \to\ (\forall x \exists y\, P(x, y)) \qquad\qquad (\neg \exists x\ x^x = x!) \to \forall y, z\ y \neq z.$$

◯ They are both false.

◯ Only the first is true.

◯ Only the second is true.

◯ They are both true.

Exercise 9. Let E be a set of endpoints on a network, let P be a set of paths connecting those endpoints, and let $C(p, x, y)$ be the proposition that path $p \in P$ connects endpoints x and y with $x, y \in E$. The statement "there are at least two paths connecting every two distinct endpoints on the network" can be expressed by

- ◯ $\forall x, y \in E \Big(x \neq y \rightarrow \exists p, q \in P \left(p \neq q \wedge (C(p, x, y) \vee C(q, x, y)) \right) \Big)$.
- ◯ $\forall x, y \in E \Big(x \neq y \wedge \exists p, q \in P \left(p \neq q \wedge C(p, x, y) \wedge C(q, x, y) \right) \Big)$.
- ◯ $\neg \Big(\exists x, y \in E \left(x \neq y \ \wedge \ \forall p, q \in P \left(p = q \vee \neg C(p, x, y) \vee \neg C(q, x, y) \right) \right) \Big)$.
- ◯ $\neg \Big(\exists x, y \in E \left(x \neq y \ \wedge \ \forall p, q \in P \left(p = q \wedge \neg C(p, x, y) \wedge \neg C(q, x, y) \right) \right) \Big)$.

Exercise 10. Given the propositional function $T(x)$, the statement $\exists! x \, T(x)$ is logically equivalent to

- ◯ $\neg(\forall x \, [T(x) \rightarrow \exists y \neq x \, T(y)])$.
- ◯ $\exists x \forall y((\neg T(y)) \vee (y = x))$.
- ◯ $\exists x(T(x) \vee \forall y[(\neg T(y)) \vee (y = x)])$.
- ◯ $\exists x(T(x) \wedge \forall y[T(y) \wedge (y = x)])$.

Exercise 11. Given the propositional functions $G(x)$: "x is a boy", $F(y)$: "y is a girl", and $A(z)$: "z likes computers", the statement "all boys like computers and there is a girl that does not like computers" can be expressed by

- ◯ $\neg[(\exists y \, (G(y) \wedge \neg A(y))) \vee (\forall x \, (F(x) \rightarrow A(x)))]$.
- ◯ $\neg[(\forall x \, (F(x) \rightarrow A(x))) \vee (\exists y \, (G(y) \rightarrow \neg A(y)))]$.
- ◯ $(\exists x \, (F(x) \rightarrow \neg A(x)) \wedge (\forall y \, ((\neg G(y)) \vee A(y)))$.
- ◯ $(\forall y \, (G(y) \wedge A(y))) \wedge (\exists x \, (F(x) \wedge \neg A(x)))$.